

(a) Unpaired Preference Data

source models



y_1



y_2

y_3

y_4

$\vdots$

y_K

$\vdots$

target model



Independently labeled desirable and undesirable completions

Desirable

D_x

y_1

y_3

$\vdots$

Undesirable

U_x

y_2

y_4

$\vdots$

(b) Source-Consensus Value Guidance

Compute source consensus $C(y|x)$

M_1

$p_1(y|x)$

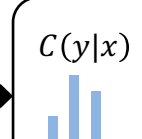
M_2

$p_2(y|x)$

$\vdots$

M_N

$p_N(y|x)$



High consensus
→ stronger source support
→ learn more

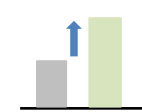
Low consensus
→ uncertain
→ learn less

Source-guided value for each completion

$$v(y|x) = \log p_\theta(y|x) - \log C(y|x)$$

Desirable ($y \in D_x$)

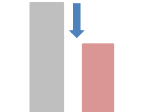
Increase r_θ^D



Preserve and reinforce verified behavior

Undesirable ($y \in U_x$)

Decrease r_θ^U

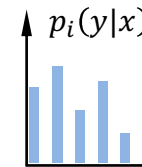


Consensus-guided error suppression

(c) Consensus Calibration and Stabilization

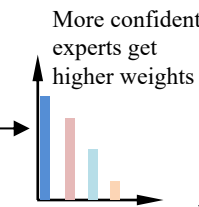
① Confidence calibration via entropy

Estimate expert confidence with predictive entropy



Entropy $H_i(y)$

Normalize & Calibrate



② Source-target baseline shift

Instance-wise normalization for fair comparison

Target baseline

$$\bar{p}_\theta(y|x) = \log p_\theta(y|x) - \mathbb{E}_{y'}[\log p_\theta(y'|x)]$$

Source baseline

$$\bar{C}(y|x) = \log C(y|x) - \mathbb{E}_{y'}[\log C(y'|x)]$$

③ D-first / U-warmup curriculum

Optimize desirable samples first, then gradually introduce undesirable samples

